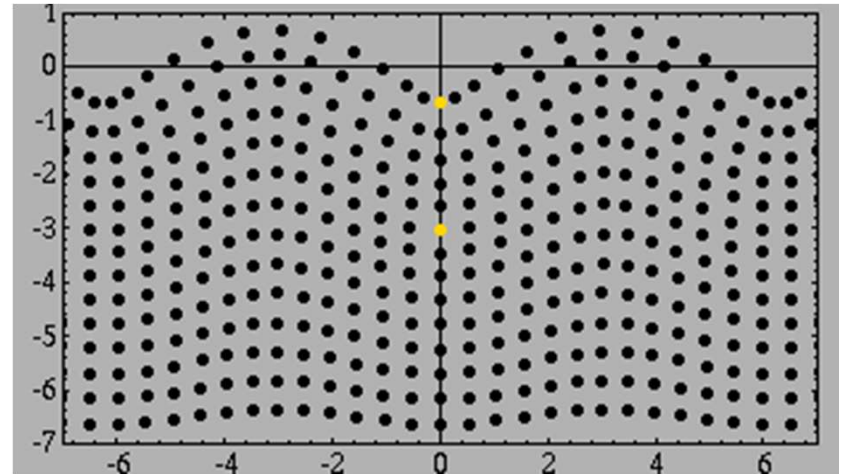


United International University
PHY 101
**Waves and Oscillation,
Optics and Modern Physics**

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Institute of Natural Sciences





Part I

Simple harmonic oscillators

Damped oscillators

Driven oscillators

Resonance

The Wave Equation



Part II

Theories of Light
Interference of Light
Diffraction of Light
Polarization of Light



Part III

Relativity

Particle Properties of Wave

Wave Properties of Particles

Atomic Physics

Nuclear Physics

What to do in this course:

1. Read the relevant sections in the textbook ... the course notes will guide you.
2. Do all homework and problem sets.
3. Get help early ... from Course Teacher.
4. there are no shortcuts ... put effort in to understand things ...

Problem-solving and homework

Each Chapter you will be given a take-home problem set to complete and hand in for marks ... In addition to this, you need to work through the problems in *Book*, in your own time, at home.

You will not be asked to hand these in for marks. Get help from your friends, the course Teacher ...

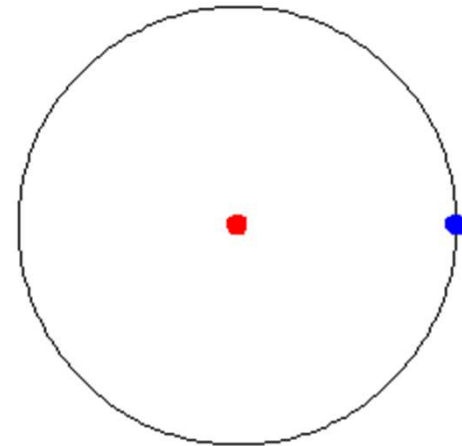
Do not take shortcuts. Mastering these problems is a fundamental aspect of this course.

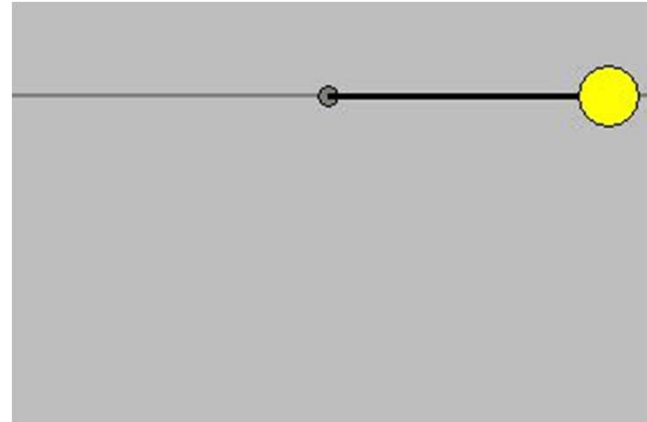
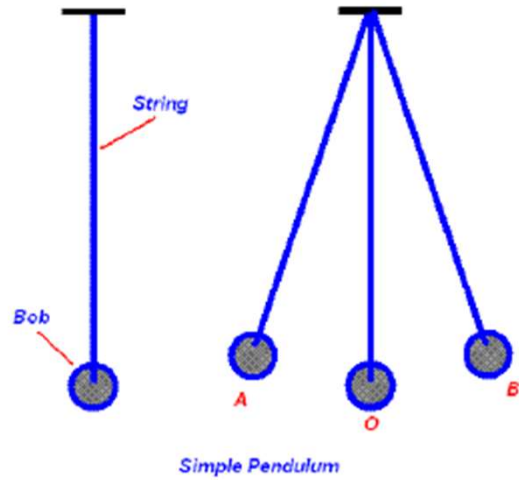
Different Kind of Motions

Linear Motion



Uniform Circular Motion





Oscillatory Motion (Simple Pendulum)



Oscillatory Motion (Spring Mass)

Oscillatory Phenomena

... observed in many physical systems ...
from the very small...(e.g. dipole resonance in nuclei)... to the very large (earthquake waves, stars,...)

Mechanical systems to lasers
..... from violin strings to electrical systems

Oscillatory Motion

Motion which is periodic in time, that is, motion that repeats itself in time.

Examples:

- Power line oscillates when the wind blows past it
- Earthquake oscillations move buildings

- 
- **Block attached to a spring**
 - **Motion of a swing**
 - **Motion of a pendulum**
 - **Vibrations of a stringed musical instrument**
 - **Motion of a cantilever**
 - **Oscillations of houses, bridges, ...**
 - **All clocks use simple harmonic motion**

Periodic Motion: Many kind of motion repeat over and over, such as, the vibrations of quartz crystal in a watch, swinging pendulum in a clock and back-and-forth motion of a piston in an engine. This kind of motions are called periodic motion.

Amplitude: The amplitude of the motion, denoted by A , is the maximum magnitude of displacement from the equilibrium position. It is always positive

Period: The period T , is the time required for one oscillation.

Frequency: The frequency, f , is the number of cycles in a unit time.

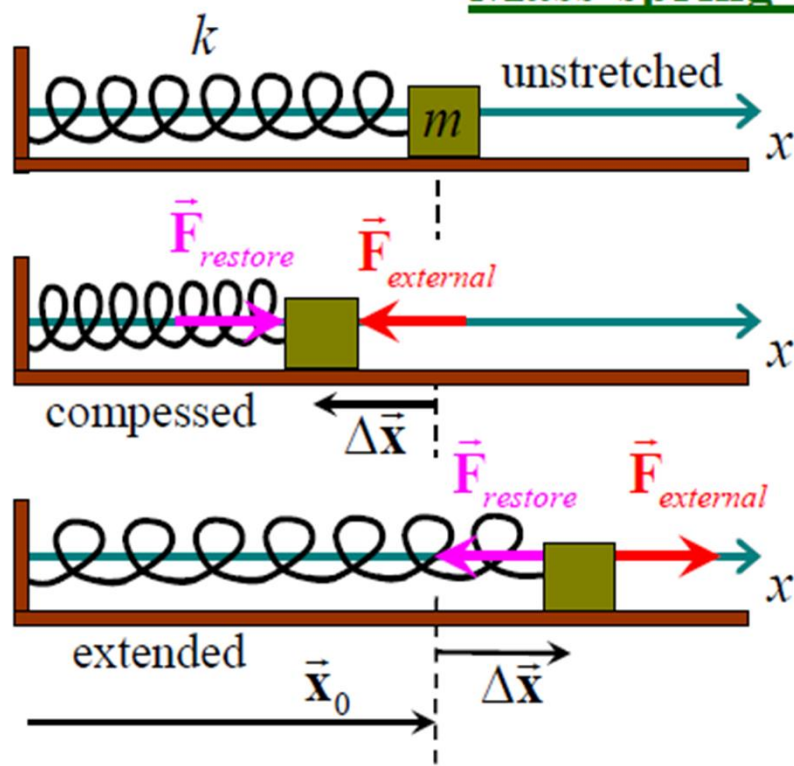
Simple Harmonic Motion: The simplest kind of oscillation occurs when the restoring force F_x is directly proportional to the displacement from the equilibrium x .

When the restoring force is directly proportional to the displacement from the equilibrium as given by equation

$$F_x = -kx$$

The oscillation is called Simple Harmonic Motion(SHM).

Mass-spring oscillator



Hooke's Law:

Restoring force,

$$\vec{F}_{\text{restore}} = -k\Delta\vec{x}$$

where $\Delta\vec{x} = \vec{x} - \vec{x}_0$

and k is the “spring constant”
[N m⁻¹]

Start with the momentum principle: $\frac{d\vec{p}}{dt} = \vec{F}_{\text{net}}$

For horizontal forces on the mass: $\frac{dp_x}{dt} = -kx$

$$\therefore \frac{d(mv_x)}{dt} = -kx \quad \text{or} \quad \frac{d}{dt} \left(m \frac{dx}{dt} \right) = -kx$$

$$\therefore \frac{d^2x}{dt^2} = -\frac{k}{m}x$$

Mass-spring oscillator ...2

$$\frac{d^2 x(t)}{dt^2} = \frac{-k}{m} x(t)$$

... a second order differential equation
... we know that if we displace a mass-spring system from its rest position and then release it, it will perform SHM ...

Guess a trial solution: $x(t) = A \cos(\omega t + \phi)$

$$\text{then } \frac{d^2 x}{dt^2} = -A\omega^2 \cos(\omega t + \phi)$$

and substitute into our DE: $-A\omega^2 \cos(\omega t + \phi) = -A \frac{k}{m} \cos(\omega t + \phi)$

... which is true provided $\omega^2 = \frac{k}{m}$

Therefore our solution is $x(t) = A \cos(\omega t + \phi)$ where $\omega = \sqrt{\frac{k}{m_{10}}}$

Mass-spring oscillator ...3

We will write a particular value of ω as ω_0 , as the **natural angular frequency** of the oscillator – the frequency that it “wants” to oscillate at.

Mass-spring system: $\omega_0 = \sqrt{\frac{k}{m}}$ and $T = 2\pi\sqrt{\frac{m}{k}}$

System parameters: $m, k \longrightarrow \omega_0$

Initial conditions: $\longrightarrow A, \phi$

(Note that ω_0 is independent of A)

Simple harmonic oscillator

$$x(t) = A \cos(\omega_0 t + \phi)$$

$$v(t) = \frac{dx(t)}{dt} = -A\omega_0 \sin(\omega_0 t + \phi)$$

$$a(t) = \frac{d^2x(t)}{dt^2} = \frac{dv(t)}{dt} = -A\omega_0^2 \cos(\omega_0 t + \phi)$$

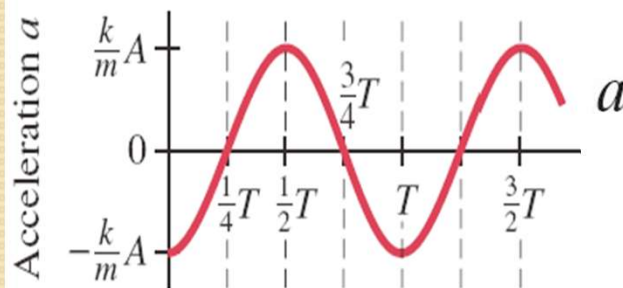
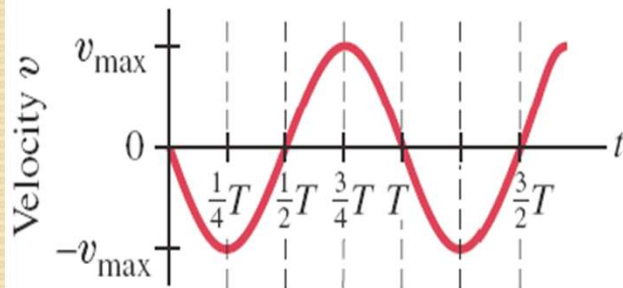
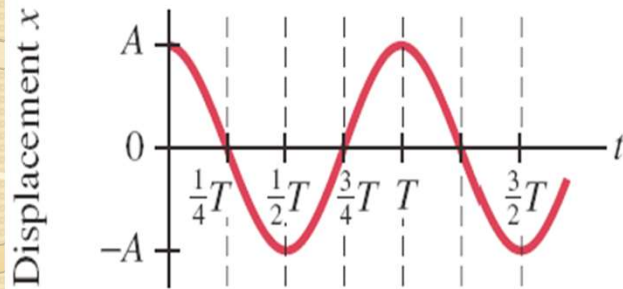
$$\dots \text{ acceleration} = -(\text{constant}) \cdot (\text{displacement})$$

$$= -A\omega_0^2 \cos(\omega_0 t + \phi)$$

$$= A\omega_0^2 \cos(\omega_0 t + \phi + \pi)$$

Phase difference between acceleration and displacement is π

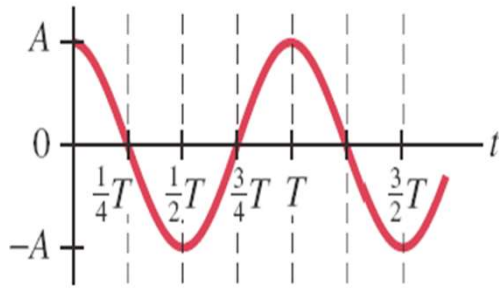
Phase difference between v and x (and v & a) is $\frac{\pi}{2}$



The velocity and acceleration for simple harmonic motion can be found by differentiating the displacement:

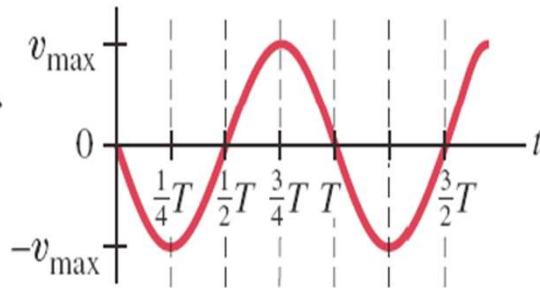
$$v = \frac{dx}{dt} = -\omega A \sin(\omega t + \phi)$$

$$a = \frac{d^2x}{dt^2} = \frac{dv}{dt} = -\omega^2 A \cos(\omega t + \phi).$$

Displacement x 

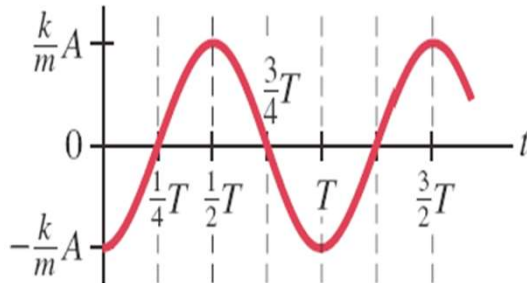
$$x = A \cos(\omega t + \phi),$$

$$v = -\omega A \sin(\omega t + \phi)$$

Velocity v 

$$= \omega A \cos(\omega t + \phi + \frac{\pi}{2}),$$

$$a = -\omega^2 A \cos(\omega t + \phi)$$

Acceleration a 

$$= -\omega^2 A \sin(\omega t + \phi + \frac{\pi}{2}).$$